

Experiments continued

How good does support-NELBO translate to downstream quality?

Stephanus Park, 04.06.2026 — Experiments continued

Agenda

What shall we talk about?

Recap

Downstream evaluation

Bottleneck hunting

Key takeaways

Stephanus Park, 04.06.2026 — Experiments continued

Recap

Where we left off last time

- Metadata and source-only proxies were useful diagnostics, but unstable routing signals
- Direct support-NELBO gave the strongest leakage-safe compatibility estimate
- New question: does support-NELBO-selected generation improve held-out downstream utility?

Stephanus Park, 04.06.2026 — Experiments continued

(Zhao et al., 2024)

3

The previous presentation ended with a positive result at the compatibility-estimation level.

Direct support-NELBO was the first method that made expert-target compatibility look measurable under the protocol.

But that result was still at the routing stage.

It told us which expert looks most compatible with a target support set, not whether the generated embeddings from that expert are useful for a downstream classifier.

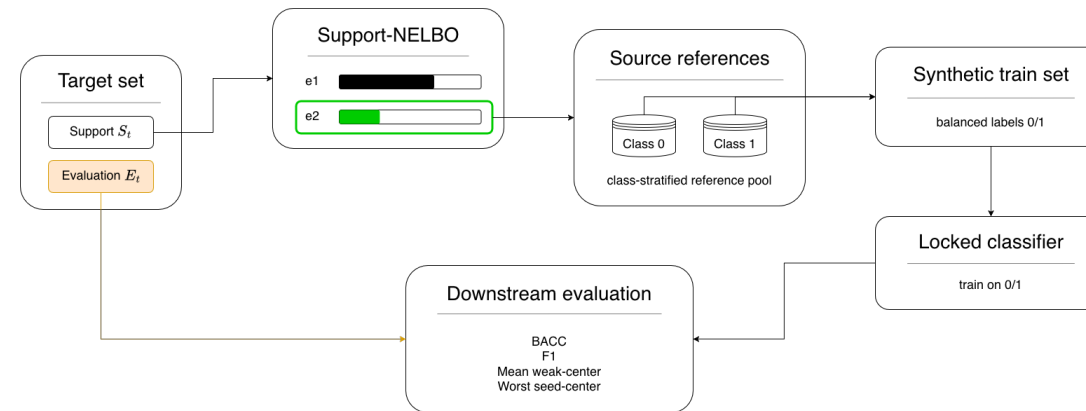
So the translation test is:

I take the support-NELBO ranking, select one or several experts, generate synthetic embeddings, train downstream classifiers, and evaluate only on the held-out target set.

This asks whether compatibility estimation survives the full downstream pipeline.

Downstream evaluation pipeline

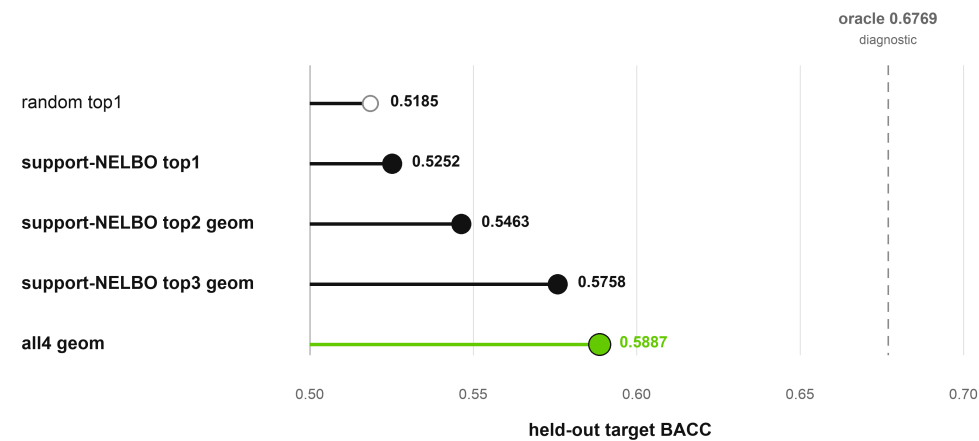
How do I evaluate downstream utility



Stephanus Park, 04.06.2026 — Experiments continued

Downstream utility evaluation

Support-NELBO on the held-out target set



Stephanus Park, 04.06.2026 — Experiments continued

5

Support-NELBO top-1 only slightly improves over random.

That means the signal that looked strong for compatibility does not directly become a reliable sparse downstream selector.

But the top-k and dense variants improve the picture.

Top-2 and top-3 geometric aggregation move upward, and all4 geometric composition reaches 0.589.

But even the diagnostic single-expert downstream oracle only reaches about 0.677 BACC.

So even perfect selection within this sparse single-expert setup would still fall below the desired utility level.

BRIDGE:

That turns the next section into a bottleneck-localization problem: where is downstream signal being lost?

Bottleneck hunting

Stephanus Park, 04.06.2026 — Experiments continued

Three possible bottlenecks

Where can downstream utility be lost?

Representation space

Are the 758-D DINOv2 embeddings too noisy for downstream utility?

Generation step

Does the CVAE preserve class-relevant geometry after generation?

Sparse routing

Is Top-1 support-NELBO selection too brittle?

Stephanus Park, 04.06.2026 — Experiments continued

7

I do not treat the low downstream result as one failure.

I split it into three possible failure points.

First, the representation may not contain enough signal.

Before testing any CVAE or router, the real embedding space must contain class-relevant and domain-relevant structure.

If DINOv2 embeddings are noisy or insufficient for histopathology, generation and routing cannot recover the lost signal.

Second, the CVAE may not preserve the useful structure during generation.

Even if the real embeddings are useful, the CVAE may fail to preserve the class-conditional geometry after sampling.

This is a legitimate bottleneck because VAEs/CVAEs optimize a variational objective, not downstream balanced accuracy directly.

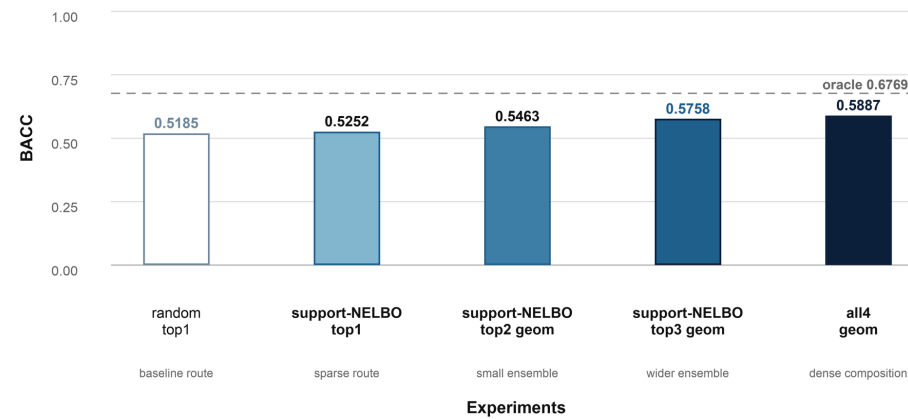
Third, the signal may exist, but sparse top-1 routing may select the wrong component.

This is exactly the sparse expert-selection problem: sparse routing is efficient, but it requires accurate routing; otherwise, useful complementary experts are discarded.

The next slides test these bottlenecks one by one.

Check 1: Representation space

758-Dimensions of DINOv2 too noisy?



Stephanus Park, 04.06.2026 — Experiments continued

(Courty et al., 2017)

8

The key point is that the 0.677 oracle does not end the analysis.

It tells me the sparse generated-expert setup has a ceiling.

To locate the ceiling, I first test whether the foundation embedding space itself is too noisy.

The PCA64 reconstruction upper reaches 0.862, so the information needed for the downstream task is still present.

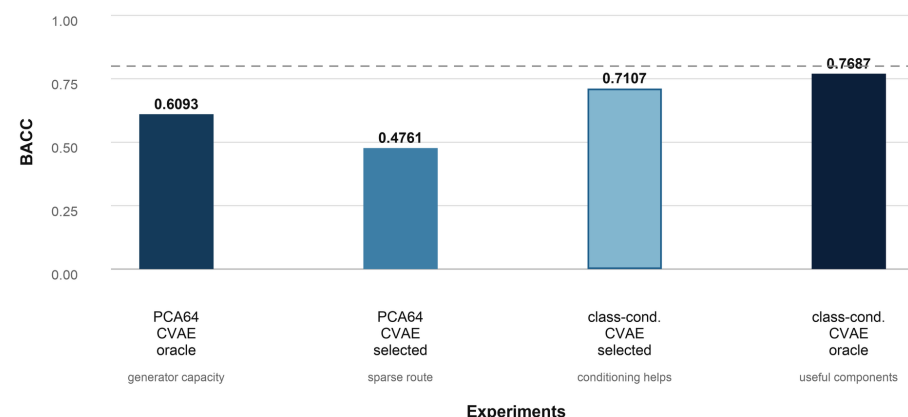
Therefore the failure is not caused by the feature representation alone; it is introduced later by CVAE generation, class preservation, or sparse expert routing.

Note:

The reported 0.862 is not the plain mean over all reconstructed experts. It is the best reconstructed-source candidate per evaluation cell, then averaged at center level.

Check 2: CVAE generation step

Do CVAE experts preserve relevant structure?



Stephanus Park, 04.06.2026 — Experiments continued

(Wen et al., 2025)
(Whitman & Latypov, 2025)

9

This slide tests the second bottleneck: whether the CVAE generation step preserves the useful structure that was still present after PCA64 reconstruction.

The first two bars show the unconditioned PCA64 CVAE.

Even the diagnostic oracle only reaches 0.609, and the selected expert drops to 0.476. So without class structure, generation loses a large part of the utility signal.

The class-conditioned pivot tested whether making the label an explicit decoder condition could preserve class geometry better.

The selected result rises to 0.711, and the oracle reaches 0.769. That tells me the generation process can preserve more useful downstream structure when class information is injected.

But this is still below the PCA64 reconstruction upper of 0.862, and still below the 0.80 utility gate. So the synthesis is: generation is a real bottleneck, but not a complete dead end.

Bridge:

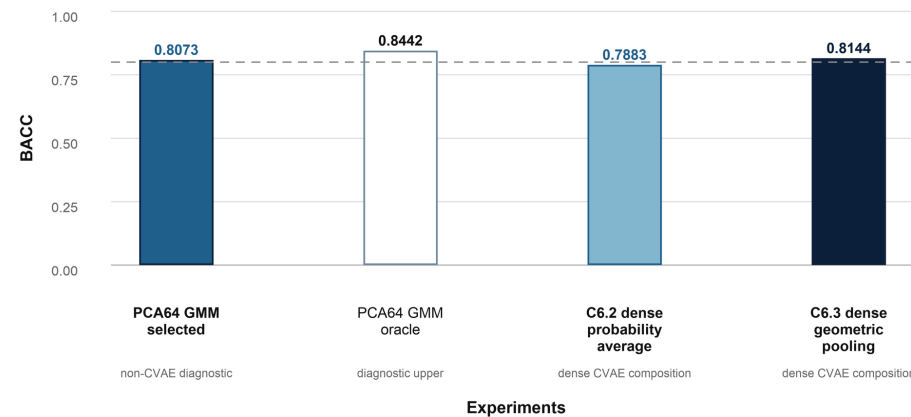
Class conditioning recovers part of the lost signal, while the remaining gap points to preservation quality and routing as the next issues.

Note:

Unconditioned PCA64 CVAE means the decoder had no explicit class condition. Class information could only enter indirectly through class-stratified source reference posterior samples, not through a decoder call of the form $d(z, y)$. The class-conditioned pivot tested whether making the label an explicit decoder condition could preserve class geometry better.

Check 3: Sparse routing

Is Top-1 selection too brittle?



Stephanus Park, 04.06.2026 — Experiments continued

(Yadav et al., 2025)

10

After the CVAE generation check, I wanted to know whether the problem is unavoidable for synthetic embeddings, or whether it is specific to the CVAE generator.

The PCA64 GMM is a simpler diagnostic sampler. It removes the CVAE encoder-decoder bottleneck and asks: if I model the source distribution more directly in PCA64 space, can synthetic embeddings support downstream classification?

The answer is yes. The selected GMM reaches 0.807, and the diagnostic oracle reaches 0.844. So the downstream pipeline itself is not doomed. A simpler sampler can preserve enough geometry to cross the 0.80 gate.

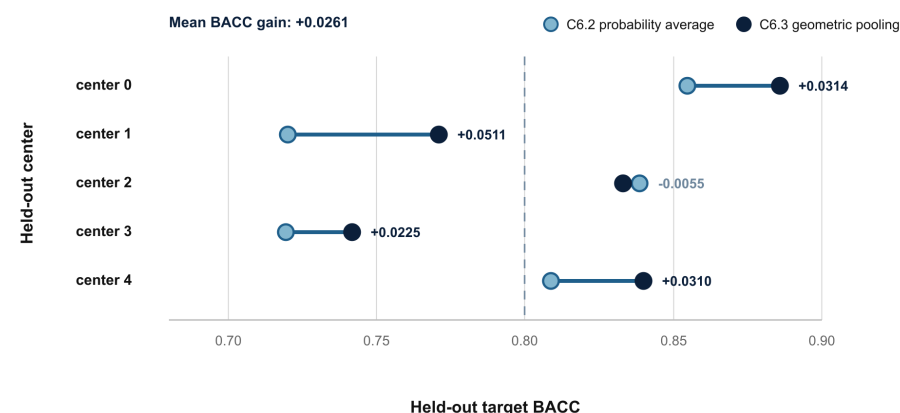
But the GMM is not the thesis method. It is a diagnostic control. It tells me that useful synthetic distributions are possible, and that the remaining CVAE problem is not only “which expert do I select”, but also how much useful structure each generated component preserves.

This motivates C6.3. Instead of trusting one sparse CVAE-generated expert, C6.3 keeps multiple frozen CVAE components, trains component-specific classifiers, and combines their probabilities with a geometric ensemble.

The key connection is: the GMM result shows that strong synthetic downstream utility is achievable when the generated distribution is clean enough. C6.3 asks whether I can recover part of that benefit while staying inside the CVAE/MoErging setting by reducing sparse-routing risk.

The result is 0.814 BACC. So the synthesis is: sparse top-1 routing is too brittle, but dense post-hoc CVAE composition can recover enough complementary signal to cross the utility gate.

Comparison of aggregation strategies



Stephanus Park, 04.06.2026 — Experiments continued

(Yadav et al., 2025)

11

This slide isolates the aggregation rule inside the dense CVAE composition result.

Up to this point, the story was: representation is not the main bottleneck, generation quality matters, and sparse top-1 routing is too brittle. C6.2 already responds to that by avoiding one selected expert: it trains component-specific classifiers and averages their probabilities.

Here, C6.3 changes only the final aggregation rule. Instead of arithmetic probability averaging, it uses geometric pooling, which is more consensus-sensitive. So this is not a new generator and not a new compatibility estimator. It asks whether dense CVAE composition should behave more like a consensus across components.

The plot shows that geometric pooling improves mean BACC by about +0.026. Four out of five centers improve: center 1 gains the most, about +0.051, and centers 0, 3, and 4 also move upward. Center 2 slightly decreases by about -0.006, so the gain is not perfectly uniform.

The synthesis is that dense composition is useful, but the aggregation rule matters. Simple averaging already reduces sparse-routing risk, but geometric pooling recovers more utility by rewarding agreement across frozen CVAE components and reducing the influence of weaker or overconfident members.

This supports the thesis-facing claim cautiously: the strongest result is not better top-1 routing, but post-hoc dense composition of independently trained CVAE experts.

Key takeaways

Stephanus Park, 04.06.2026 — Experiments continued

Stephanus Park, 04.06.2026 — Experiments continued

References

- Courty, N., Flamary, R., Tuia, D., & Rakotomamonjy, A. (2017). Optimal transport for domain adaptation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 39(9), 1853–1865. <https://doi.org/10.1109/TPAMI.2016.2615921>
- Wen, X., Zhao, B., Elezi, I., Deng, J., & Qi, X. (2025). “Principal Components” Enable A New Language of Images (Version 2). arXiv. <https://doi.org/10.48550/ARXIV.2503.08685>
- Whitman, S. E., & Latypov, M. I. (2025). Machine learning of microstructure–property relationships in materials leveraging microstructure representation from foundational vision transformers. *Acta Materialia*, 296, 121217. <https://doi.org/10.1016/j.actamat.2025.121217>
- Yadav, P., Raffel, C., Muqeeth, M., Caccia, L., Liu, H., Chen, T., Bansal, M., Choshen, L., & Sordoni, A. (2025). A Survey on Model MoErging: Recycling and Routing Among Specialized Experts for Collaborative Learning (arXiv:2408.07057). arXiv. <https://doi.org/10.48550/arXiv.2408.07057>
- Zhao, Z., Xu, Y., Zhang, J., Zhao, R., Chen, Z., & Jiao, Y. (2024). A Semi-supervised Gaussian Mixture Variational Autoencoder method for few-shot fine-grained fault diagnosis. *Neural Networks*, 178, 106482. <https://doi.org/10.1016/j.neunet.2024.106482>

Stephanus Park, 04.06.2026 — Experiments continued